VAAL UNIVERSITY OF TECHNOLOGY

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Week 1
EIOPS4A Lab Guide

Week 1: KVM Cluster Setup (Alternative)

Objective: Build a 3-node HPC cluster sandbox natively on Linux using KVM/QEMU and virt-manager.
Goal: Install the virtualization stack, configure an isolated virtual network, install the master node, and clone it to create compute nodes. This is an alternative to the Vagrant lab.

1. Prerequisites & Installation

This lab is designed for students running a native Linux host (e.g., Ubuntu, Fedora, Debian). Kernel-based Virtual Machine (KVM) is an open-source virtualization technology built directly into Linux.

Install KVM and virt-manager (Ubuntu/Debian)

```
sudo apt update
sudo apt install qemu-kvm libvirt-daemon-system libvirt-clients bridge-utils
virt-manager
sudo adduser $USER libvirt
sudo adduser $USER kvm
```

Note: You may need to log out and log back in for the group changes to take effect.

Install KVM and virt-manager (Fedora/RHEL)

```
sudo dnf install @virtualization
sudo systemctl enable --now libvirtd
sudo usermod -aG libvirt $USER
```

2. Virtual Network Setup

Our cluster needs an isolated internal network to communicate without external interference.

1. Open **Virtual Machine Manager**.
2. Go to **Edit** > **Connection Details** > **Virtual Networks**.
3. Click the **+** button to add a new network.
4. **Name:** `hpc-internal`.
5. **IPv4 configuration:** Network: `10.0.0.0/24`. Disable DHCP (we will use static IPs).
6. Finish creating the network.

3. Creating the Master Node

We will manually create and configure the Master Node, then clone it for the compute nodes.

1. Download the openEuler 24.03 ISO.
2. In `virt-manager`, click **Create a new virtual machine**.
3. Choose **Local install media (ISO image)** and browse to your downloaded openEuler ISO.
4. **Memory:** 2048 MB (2 GB). **CPU:** 2 Cores.
5. **Storage:** 20 GB.
6. **Name:** `hpc-master`. Before clicking Finish, check **Customize configuration before install**.
7. In the hardware details, you will have one NIC (Network Interface Card) attached to the default NAT network (for internet).
8. Click **Add Hardware** > **Network**. Select the `hpc-internal` network you created.
9. Click **Begin Installation**. Install openEuler, set a root password, and create a user.

4. Configuring the Master Node

Boot into your new `hpc-master` VM and log in.

```
# Set the hostname
sudo hostnamectl set-hostname master

# Configure the internal network interface (assuming enp2s0 is internal)
sudo nmcli connection add type ethernet con-name eth-internal ifname enp2s0 i
pv4.addresses 10.0.0.1/24 ipv4.method manual
sudo nmcli connection up eth-internal
```

```
# Shut down the VM to prepare for cloning
sudo poweroff
```

5. Cloning Compute Nodes

1. In `virt-manager`, right-click the powered-off `hpc-master` VM and select **Clone**.
2. **Name:** `compute-01`. Ensure you select to clone the disk.
3. Repeat the process to create `compute-02`.

6. Reconfiguring Compute Nodes

Start `compute-01` and `compute-02`. Log into each through the virt-manager console and update their identities.

On compute-01:

```
sudo hostnamectl set-hostname compute-01
sudo nmcli connection modify eth-internal ipv4.addresses 10.0.0.101/24
sudo nmcli connection up eth-internal
```

On compute-02:

```
sudo hostnamectl set-hostname compute-02
sudo nmcli connection modify eth-internal ipv4.addresses 10.0.0.102/24
sudo nmcli connection up eth-internal
```

7. Verification & Submission

Start your Master Node. Verify cluster connectivity:

```
ping -c 3 10.0.0.101
ping -c 3 10.0.0.102
```

Submit a screenshot showing the `virt-manager` main window with all 3 VMs running, alongside your master node terminal successfully pinging the compute nodes. Save this screenshot as `Week1_Lab_Surname.png`.